Jianxuan Shao

Sydney, Australia | jsha0336@uni.sydney.edu.au | Portfolio | Game Fellows

Research Interests

I want to build personal AI companions that follow what a person is doing, reason about what help is appropriate, and adapt through repeated collaboration. I'm interested in representing what an agent knows about a task and how it should act when observations are incomplete or corrected.

Education

University of Sydney

Feb 2025 to present

Master of Computer Science | Data Science and AI | Expected completion: Dec 2026

Average across 12 graded units: 79.33/100. Semester 1, 2026 average: 84.5/100.

Selected results: Machine Learning and Data Mining 89/100; Introduction to Programming 87/100; Computer and Network Organisation 85/100.

Hebei University

Sep 2018 to Jun 2023

Bachelor of Engineering in Architecture | Five-year programme | Average: 86.0/100

Hebei University Scholarships: Second Prize (2019); Third Prize (2021, 2022).

Research and Projects

Game Fellows

2025 to present

Independent prototyping and supervised research development | Python and Pygame

- Built a playable companion that follows formations, intercepts projectiles and coordinates support fire using game-state rules. Separately explored using successive screenshots to infer game state and inform actions and spoken feedback.
- Implemented event logging and designed counterbalanced solo and companion conditions with paired random seeds. Prepared measures of teammate presence, trust and player control, and developed a manuscript under academic guidance.

Explainable AI Generated Image Detection

2026

Master's capstone | Research design and error analysis in an eight-person team

- Audited 60,000 predictions across 10 model runs against the dataset manifest. Independently reconstructed 900 fusion decisions from frozen outputs and analysed disagreements between a vision-language model and a visual forensics model.
- Categorised 51 errors on an internal split used during model screening and documented cases where fusion introduced new errors.

Additional Experience

Damo Cave Immersive Exhibition

Jul 2023 to Jul 2024

Interactive Media Intern | Shanghai Fengyuzhu Culture Technology

Helped integrate and debug multichannel projection, wall-touch controls and radar sensing on site.

Technical Skills

Python, SQL, Pygame, event logging, experiment design, model evaluation and reproducible error analysis; JavaScript and Three.js for browser prototypes.